12-02 第1篇

阅读理解（说明文） Why Don't Fish Freeze in an Icy Winter Lake?

① Every December, if you take a careful walk along a northern Chinese lake — at Daming Lake in Jinan, at Songhua Lake near Jilin City, or at small village ponds up in Liaoning — you will notice something quietly surprising. The **whole top** of the lake freezes: a smooth, hard, white-grey sheet of ice, often thick enough for children to skate on. But a week later, when ice-fishermen saw a small hole in the sheet, you can see something even more surprising. **Perch, carp, and crucian carp are swimming there, slowly but alive**, as if the whole winter were only a quiet dream. How can fish live through months in a lake that, on top, looks like one large block of frozen water? The answer rests on three quiet pieces of physics and one quiet piece of biology.

② **Piece one: water is unusual.** Most substances get denser the colder they get. Water does too — but only down to about **4 ℃**. Below 4 ℃, water starts to get **lighter again**, not heavier. At 0 ℃, ice is about **9% lighter** than liquid water at 4 ℃. That is why ice floats on the top of your glass; and, much more importantly, why a sheet of ice floats on the top of the lake rather than sinking to the bottom. If water behaved like most liquids — heaviest at its freezing point — lakes would freeze **from the bottom up**, and every fish would be locked into ice within a single winter. The chemistry of the water molecule, with its famous "V" shape, is quietly the reason lake life is possible at all in cold places.

③ **Piece two: the lake keeps its warmest water at the bottom.** Because water is densest at 4 ℃, once winter starts, the coldest layer (0—2 ℃) rises to the top and touches the ice. The **warmer, denser 4 ℃ water sinks and stays near the lake floor** — forming a quiet, sheltered layer where most fish spend the winter. A 2023 study of 22 winter lakes in Heilongjiang measured the lake-floor temperature throughout January and February. The surface ice was about −4 ℃; the water just below the ice, close to 0 ℃; but three metres down, **a steady 3.4—4.1 ℃** — cool enough to slow fish down, warm enough never to freeze them solid. The fish do not have to work to keep this layer; the lake, by the laws of density alone, arranges it for them.

④ **Piece three: ice is an excellent blanket.** Ice looks cold, but it is a surprisingly good **thermal insulator**. Once a sheet of ice forms on the lake's top, it slows the further loss of heat from the water below into the freezing air above. In a 2021 Swedish winter study, a 30-centimetre ice cover reduced heat loss from the lake water beneath by about **85%** compared to open water in the same wind. This is why, even in a −25 ℃ January, the water **one metre below the ice** in a big lake rarely drops below 1 ℃. The ice, once it is there, protects exactly the world it seems to be imprisoning.

⑤ **The biological piece: fish slow down on purpose.** Unlike us, fish are **cold-blooded**: their body temperature tracks the water. In summer, a carp's heart beats around **60—80 times per minute**; in a 4 °C winter lake, it beats only **6—10 times**. Its whole metabolism — breathing, digesting, swimming, growing — drops to about one-tenth of its summer level. This state, called **winter torpor**, lets fish live for months on almost nothing. Many do not eat for ten or twelve weeks; they lose only a little weight because almost no energy leaves their bodies. They hang quietly near the lake floor, eyes open, fins barely moving, like small grey leaves that remembered how to be alive.

⑥ What about **oxygen**? This is the real winter risk, not the cold. In a sealed, ice-covered lake, oxygen cannot enter from the air. Over a long winter, fish can slowly use up the dissolved oxygen, especially if there is also rotting plant matter below. In most healthy lakes, this is fine — winter lakes hold **more** dissolved oxygen than summer lakes (cold water dissolves gases better), so one or two months are safe. But in very thick ice, after heavy snow (which blocks under-ice photosynthesis by the few plants still alive), "winterkill" sometimes happens. Modern lake management in northern China, since 2019, includes **drilling air holes** in long winters, and even installing small circulation pumps in 120 important lakes — a small, humane help, nothing more.

⑦ A few fish species do **something different**. **Crucian carp**, a Chinese village-pond favourite, can survive up to **four months in almost no oxygen** by turning sugar into ethanol in the muscles — essentially "breathing" without oxygen. **Arctic cod** and some polar fish make **antifreeze proteins** that bind to ice crystals and stop them from growing inside the fish's body. But most of our common lake fish — carp, crucian, perch, pike — do not need either trick. They simply sleep, shallowly and patiently, on a lake floor kept at 4 °C by the quiet geometry of water molecules.

⑧ So the next time you walk on the top of a frozen lake, and the children around you are laughing and skating, remember what is happening three metres below. The heaviest water in the world sits there, at 4 °C, holding a forest of carp and perch **absolutely still**, their hearts beating eight times a minute, waiting. The ice above is not a lid. It is, for them, **a roof, a blanket, and a long gentle winter night**. When the first thaw comes in March and the ice cracks like a distant piano, they will rise. **They have been, quietly, alive the whole time.**

注释： torpor 蛰伏（冬眠式的代谢降低）；thermal insulator 隔热体；metabolism 新陈代谢；winterkill 冰封窒息；ethanol 乙醇；antifreeze protein 抗冻蛋白

根据短文内容，选择最佳答案。

1. According to the passage, at what temperature does water reach its maximum density?

A. At 100 ˚C.
B. At 0 ˚C, the freezing point.
C. At about 4 ˚C, because below 4 ˚C water starts to get lighter again, not heavier; at 0 ˚C ice is about 9% lighter than liquid water at 4 ˚C, which is why ice floats on top.
D. At −4 ˚C.

2. What did the 2023 study of 22 winter lakes in Heilongjiang find about the lake floor temperature in January and February?

A. It was always below 0 ˚C.
B. The surface ice was about −4 ˚C, the water just below the ice close to 0 ˚C, but three metres down, a steady 3.4–4.1 ˚C — cool enough to slow fish down, warm enough never to freeze them solid.
C. The floor was warmer than summer.
D. The floor was always exactly 0 ˚C.

3. Why is ice called "an excellent blanket" in Paragraph ④?

A. Because it looks white like a blanket.
B. Because, according to a 2021 Swedish winter study, a 30-cm ice cover reduced heat loss from the lake water beneath by about 85% compared to open water in the same wind, so the water one metre below the ice rarely drops below 1 ˚C even at −25 ˚C in January.
C. Because it is very soft.
D. Because people put blankets on top of it.

4. What special ability does the crucian carp have, according to Paragraph ⑦?

A. It grows fur in winter.
B. It leaves the lake and walks on ice.
C. It can survive up to four months in almost no oxygen by turning sugar into ethanol in the muscles, essentially "breathing" without oxygen.
D. It sleeps above the ice.

5. What is the passage's main idea?

A. Fish survive an icy winter lake because water reaches its maximum density at 4 ˚C (so ice floats and the lake floor stays around 3–4 ˚C), because ice is an effective insulating blanket, because fish drop into winter torpor at one-tenth of summer metabolism, and because dissolved oxygen in cold lakes is usually enough for the long, quiet winter.
B. Ice is dangerous and fish always die.
C. Fish should be moved to warm tanks in winter.
D. Fish can fly to the south.

答案与解析

1.
【答案】C
【题型】细节理解题
【定位】第②段"4 ℃ 密度最大"
【解析】"9% 轻 / 漂浮"直引。A、B、D 均错。故选 C。

2.
【答案】B
【题型】细节理解题
【定位】第③段黑龙江研究
【解析】"−4 / 0 / 3.4−4.1"直引。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段瑞典研究
【解析】"85% / −25 / 1 ℃"直引。A、C、D 均错。故选 B。

4.
【答案】C
【题型】细节理解题
【定位】第⑦段鲫鱼段
【解析】"四个月 / 乙醇"直引。A、B、D 均错。故选 C。

5.
【答案】A
【题型】主旨大意题
【定位】全文三物理 + 一生物 + 氧气
【解析】A 对全文概括。B、C、D 均偏离。故选 A。